



"Politehnica" University of Timisoara



# Web Programming

C6. Web Multimedia Technologies.

Web CMS. SMIL. SVG



# Web Multimedia Applications

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- Combines various types of multimedia data:
  - Text
  - Images
  - Sounds
  - Video
- Various technologies involved:
  - HTML + CSS
  - Web CMS, SMIL, SVG ...
  - Client Side Scripting + Server Side Scripting + Ajax



# Web CMS

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- **Content Management System** -  
Collection of procedures used to manage workflow in a collaborative environment
- **CMS data**: documents, movies, pictures, phone numbers, scientific data etc.
- **CMS** used for storing, controlling, revising, semantically enriching, and publishing multimedia documents



# Joomla

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- **Joomla** is an open-source Web based Content Management System (WCMS)
- Joomla manages and displays user content (organized into sections, categories and articles) using page templates
- Joomla Application Framework: used to create add-ons that extend Joomla using PHP



# Joomla Article Manager

**Joomla! Administration** Version 1.5.0

Site Menus Content Components Extensions Tools Help Preview 0 1 Logout

## Article Manager

Unarchive Archive Publish Unpublish Move Copy Trash Edit New Parameters Help

Filter: [ ] [Go] [Reset] - Select Section - - Select Category - - Select Author - - Select State -

| #  | <input type="checkbox"/> | Title                                                 | Published | Front Page | Order | Access Level | Section       | Category      | Author        | Date     | Hits | ID |
|----|--------------------------|-------------------------------------------------------|-----------|------------|-------|--------------|---------------|---------------|---------------|----------|------|----|
| 1  | <input type="checkbox"/> | Example Pages and Menu Links                          |           |            | 1     | Public       |               |               | Administrator | 12.10.06 | 42   | 43 |
| 2  | <input type="checkbox"/> | What's New In 1.5?                                    |           |            | 1     | Public       | About Joomla! | The CMS       | Administrator | 11.10.06 | 88   | 22 |
| 3  | <input type="checkbox"/> | Joomla! Overview                                      |           |            | 2     | Public       | About Joomla! | The CMS       | Administrator | 09.10.06 | 146  | 19 |
| 4  | <input type="checkbox"/> | Extensions                                            |           |            | 3     | Public       | About Joomla! | The CMS       | Administrator | 11.10.06 | 100  | 26 |
| 5  | <input type="checkbox"/> | Joomla! Features                                      |           |            | 4     | Public       | About Joomla! | The CMS       | Administrator | 08.10.06 | 59   | 18 |
| 6  | <input type="checkbox"/> | Content Layouts                                       |           |            | 5     | Public       | About Joomla! | The CMS       | Administrator | 12.10.06 | 70   | 24 |
| 7  | <input type="checkbox"/> | Joomla! Facts                                         |           |            | 1     | Public       | About Joomla! | The Community | Administrator | 09.10.06 | 50   | 21 |
| 8  | <input type="checkbox"/> | The Joomla! Community                                 |           |            | 2     | Public       | About Joomla! | The Community | Administrator | 12.10.06 | 51   | 27 |
| 9  | <input type="checkbox"/> | Support and Documentation                             |           |            | 1     | Public       | About Joomla! | The Project   | Administrator | 09.10.06 | 6    | 20 |
| 10 | <input type="checkbox"/> | Joomla! License Guidelines                            |           |            | 2     | Public       | About Joomla! | The Project   | Administrator | 20.08.04 | 98   | 5  |
| 11 | <input type="checkbox"/> | Platforms and Open Standards                          |           |            | 3     | Public       | About Joomla! | The Project   | Administrator | 11.10.06 | 11   | 23 |
| 12 | <input type="checkbox"/> | Where did the Installers go?                          |           |            | 1     | Public       | FAQs          | Current Users | Administrator | 10.10.06 | 3    | 36 |
| 13 | <input type="checkbox"/> | What happened to the locale setting?                  |           |            | 2     | Public       | FAQs          | Current Users | Administrator | 06.10.06 | 10   | 13 |
| 14 | <input type="checkbox"/> | How do I upgrade to Joomla! 1.5 ?                     |           |            | 3     | Public       | FAQs          | Current Users | Administrator | 30.09.06 | 13   | 11 |
| 15 | <input type="checkbox"/> | Only one edit window! How do I create "Read more..."? |           |            | 4     | Public       | FAQs          | Current Users | Administrator | 06.10.06 | 19   | 16 |

Display # 15 [Start] [Prev] 1 2 3 [Next] [End] Page 1 of 3

Published, but is Pending | Published and is Current | Published, but has Expired | Not Published | Archived

Click on icon to toggle state.

Joomla! is Free Software released under the GNU/GPL License.

www.versiontracker.com



# Joomla development

The screenshot displays the Joomla! 4.0.0-dev Control Panel. The interface is divided into a left sidebar with navigation links and a main content area with various modules.

**Left Sidebar Navigation:**

- System
- Users
- Menus
- Media
- Content
- Components
- Extensions
- Help

**Control Panel Header:** Joomla! 4.0.0-dev

**Main Content Area Modules:**

- New Article** (Icon: Pencil)
- Media** (Icon: Image)
- Global** (Icon: Gear)
- Modules** (Icon: Cube)
- Joomla is up to date.** (Green status box)
- Extensions are up to date.** (Green status box)

**Recently Added Articles:**

| Article Title        | Date             |
|----------------------|------------------|
| Popular Tags         | 2013-10-31 00:43 |
| Similar Tags         | 2013-10-31 00:14 |
| Webbegone            | 2011-01-01 00:00 |
| Related Items Module | 2011-01-01 00:00 |
| Footer Module        | 2011-01-01 00:00 |

**Popular Articles:**

| Article Title         | Date             |
|-----------------------|------------------|
| 74 Joomla! Testing    | 2011-01-01 00:00 |
| 5 Similar Tags        | 2013-10-31 00:14 |
| 1 Popular Tags        | 2013-10-31 00:43 |
| 0 Most Read Content   | 2011-01-01 00:00 |
| 0 Latest Users Module | 2011-01-01 00:00 |

**Statistics:**

|                  |
|------------------|
| PHP 7.1.8        |
| MySQL 5.6.35     |
| Caching Disabled |
| Gzip Disabled    |
| Users 1          |
| Articles 68      |

**Logged-in Users:**

| User       | Role           | Date             |
|------------|----------------|------------------|
| Super User | Administration | 2017-11-30 10:00 |

**Custom Module:**

Example custom module for the testing data.



Joomla!®

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Log in

Register

Find the extension you're looking for

- Slideshow

Paid download

Smart Slider

By NEXTEND

SLIDESHOW

Smart Slider 3 by Nextend is the perfect slider solution for Joomla, that contains 4 slider types and is totally responsive. Create beautiful sliders & animated blocks with the next generation visual editor. Awesome

★ 5

286 reviews

J3

J4 BETA

Free

Slideshow CK

By Cédric KEIFLIN

SLIDESHOW

Slideshow CK can display images or videos with links with nice effects. It is mobile compatible and responsive design, and you can add unlimited slides with a Drag & Drop admin interface. List of features :

★ 5

180 reviews

J3

J4 ALPHA

Paid download

Unite Slider

By Unite Cms

SLIDESHOW

Unite Slider is an innovative, responsive Joomla Slider Extension that displays your content in the most beautiful way. Whether it's a Slider, Carousel, Hero Scene or even a whole Front Page, the visual drag &

★ 5

133 reviews

J3

Paid download

Creative Slider

By Ofajin

SLIDESHOW

Creative Slider is a premium multi-purpose layer slider for creating image gallery sliders, content sliders, and mind-blowing slideshow with must-see effects. This slideshow uses cutting edge technologies to provide

★ 5

97 reviews

J3

Free

Creative Image Slider

By Creative Solutions

SLIDESHOW

Creative Image Slider is a responsive image slider with amazing visual effects. It uses horizontal scrolling to make the slider more creative and attractive. It is packed with a live-preview wizard to create fantastic

★ 5

42 reviews

J3

Paid download

Unite Nivo Slider PRO

By Unite Cms

SLIDESHOW

1 Responsive Slider for Joomla. More than 100 custom graphical arrows More than 100 bullets Drag & drop control interface Dynamic images resizing Drag slider to re-size What you get is what you see interface Very

★ 5

24 reviews

J3

Dan Pescaru – Web Programming. C6

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# Joomla Resources

- Joomla Website

<http://www.joomla.org>

- Joomla Tutorial

<https://docs.joomla.org/Tutorials:Beginners>

- Free Joomla templates

<https://nicepage.com/joomla-templates>

Launch.joomla.org  
How does it work?

Your fully functional Joomla website will be hosted on the CloudKitchens.net Platform. It even includes FTP, MailboxAdmin, and the ability to install extensions. You can keep it as long as you login and click a "renew" button once every 30 days. The only limitations are 500MB of disk space, subdomain usage, and your imagination... but fear not, you can upgrade to remove the first two limitations starting at only \$5/mo.

[Find out more](#)







## Other CMSs

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- **WordPress** – open-source Web CMS written in PHP with thousands of themes and plugins available 
- **Content Hub** – paid (subscription), for marketers and business owners, includes a WYSIWYG editor and powerful built-in marketing automation tools 
- **Drupal** – open-source, best for highly customized sites, used by 71% of the Top 100 Universities 



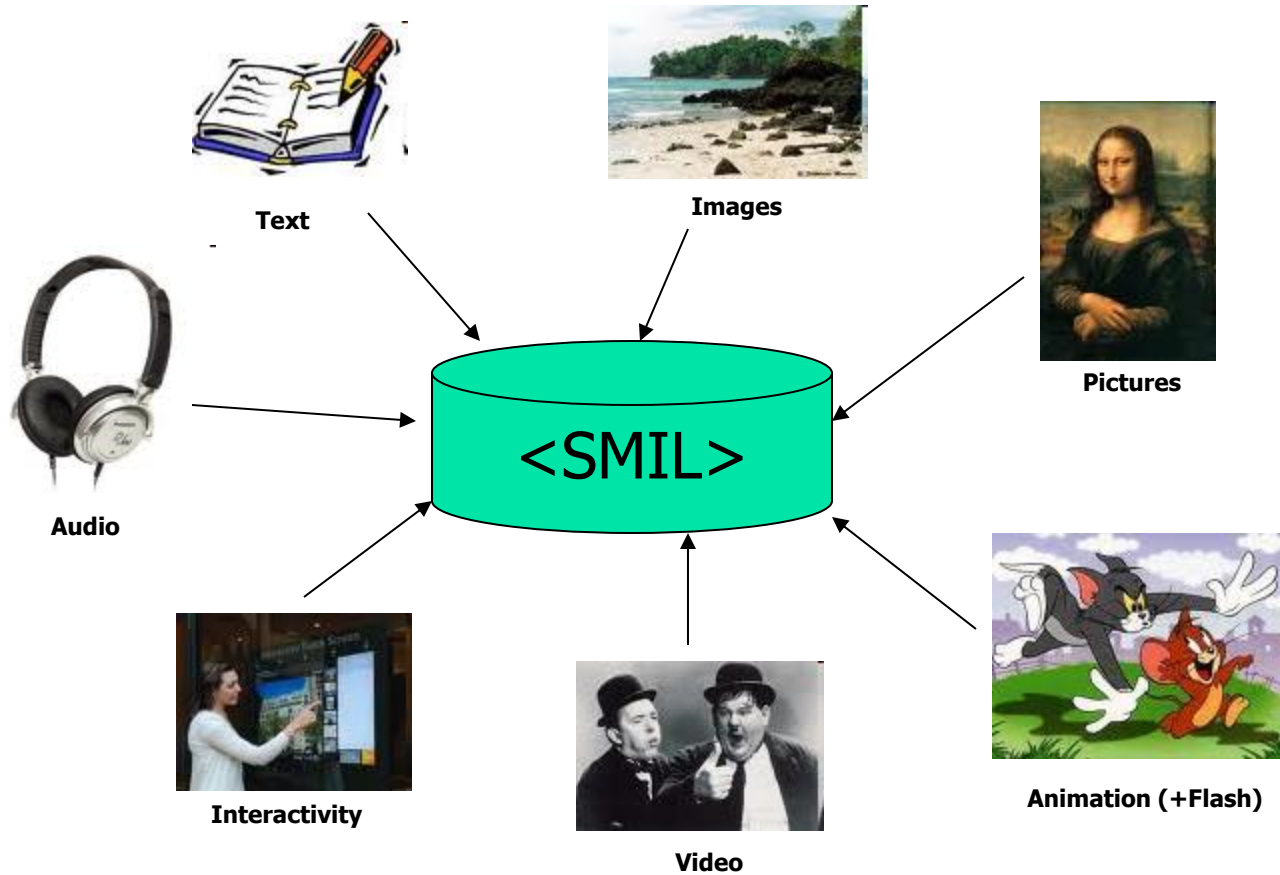
# SMIL

---

- Synchronized Multimedia Integration Language
- W3C standard
  - **SMIL** Animation (light set without time containers)
  - **SMIL** v. 3.0 language (latest standard)
- Mark-up for:
  - Timing / Layout
  - Animations / Visual transitions
  - Media embedding ...



# SMIL - Overview





# SMIL Language (I)

---

- **SMIL** documents are similar in structure to HTML documents
- Two main sections: **head** and **body**
- **<head>**
  - Layout and Meta information
- **<body>**
  - **<seq>** - sequential playlists
  - **<par>** - multiple layers in parallel
  - **<excl>** - exclusives events triggered by interrupts



# SMIL Language (II)

---

- Code in lowercase (both tags and attributes)
- All tags must be closed
- Attribute values must be enclosed in double quotes
- HTML-style comments  
*<!-- comments -->*



# SMIL 3.0 Structure

---

- Modularization breaks the language into modules
  - One module has only a small set of elements
  - Reusable element packages
- Profiling combines modules into usable languages
  - Small profiles for small devices
  - Complex profiles for big machines



# SMIL Module Groups

---

- **Animation** —placing animation
- **Content Control** —presentation flow and format
- **Layout** — additional support for layout
- **Linking** — hyperlinking
- **Media Object**— manages media objects
- **Metainformation**— extra<meta> tags
- **Structure**— greater presentation control
- **Timing and Synchronization**—more timeline control
- **Time Manipulations**— time characteristics as the speed or rate of a media object
- **Transition Effects**—transition between media objects





# Layouts (I)

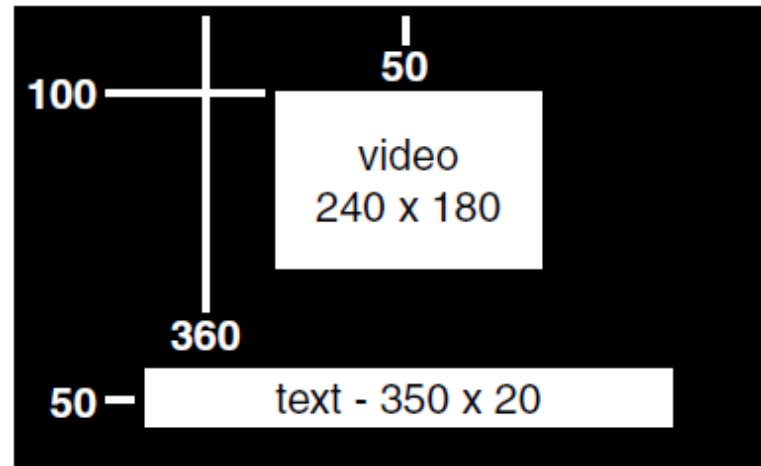
---

- Defined by the **<layout>** tag inside the **<head>** section
- Determines how objects are displayed in the presentation space
- **<root-layout>** - defines dimensions and background for the initial presentation window  
`<root-layout backgroundColor="blue"  
width="200" height="250"/>`
- **<region>** - defines playback areas



# Layouts (II)

```
<layout>
  <region id="video" top="50px" left="100px" width="240px" height="180px"/>
  <region id="text" top="360px" left="50px" width="350px" height="20px"/>
</layout>
```



(Ex. from: T. Kenedy and M. Slowinsky: "Smil: Adding multimedia to the Web")



## Layouts (III)

---

- **<regPoint>** - defines a point on the presentation display surface that is used to align elements, relative to the upper-left corner of its parent layout

```
<regPoint id="topMargin" top="10" left="15"/>
```

- **<topLayout>** - multiple-windows alternative to **<root-layout>**. Control is provided for opening and closing of top-level windows (orthogonal with presentation timeline)



# Synchronization (I)

---

- **<seq>** - plays each child element sequentially
- **<par>** - plays child elements in parallel
- **<excl>** - allows only one child element to be played at a time, triggered by events
- **Attributes:** begin, dur, end, endsync, fill, fillDefault, higher, lower, max, min, pauseDisplay, peers, repeat, repeatCount, repeatDur, restart, restartDefault, syncBehavior, syncBehaviorDefault, syncMaster, syncTolerance, syncToleranceDefault, timeAction



# Synchronization (II)

---

```
<par>
  <seq>
    
    
    
  </seq>
  <switch>
    <audio src="french.wav" system-language="fr"/>
    <audio src="english.wav" system-language="en"/>
  </switch>
</par>
```



# Media Elements

---

`<video region="r1" src="video.avi"/>`  
``,  
`<audio src="http://server/sound.wav"/>`,  
`<text src="text.html"/>`,  
`<textstream />`  
`<animation />`  
`<ref />` - for other media types



# Transitions

- `<transition>` tag. Attributes: id, type, subtype, dur, startProgress, endProgress, direction
- Types: Edge Wipes, Iris Wipes, Clock Wipes, Matrix Wipes, Push Wipe, Slide Wipe, Fade

```
<head>
  <transition id="trans1" type="barWipe" />
</head>
<body>
  <seq>
    <ref id="e1" . . . />
    <ref id="e2" transIn="trans1" . . . />
  </seq>
</body>
```



# Edge Wipes Transitions

|         |                       |                                                                                     |                |                       |                                                                                       |                  |                          |                                                                                       |
|---------|-----------------------|-------------------------------------------------------------------------------------|----------------|-----------------------|---------------------------------------------------------------------------------------|------------------|--------------------------|---------------------------------------------------------------------------------------|
| barWipe | leftToRight (default) |    | fourBoxWipe    | cornersIn (default)   |    | diagonalWipe     | topRight                 |    |
|         | topToBottom           |    |                | cornersOut            |    |                  |                          |                                                                                       |
| boxWipe | topLeft (default)     |    | barnDoorWipe   | Vertical (default)    |    | bowTieWipe       | vertical (default)       |    |
|         | topRight              |    |                | horizontal            |    |                  | horizontal               |    |
|         | bottomRight           |    |                | diagonalBottomLeft    |    | miscDiagonalWipe | doubleBarnDoor (default) |    |
|         | bottomLeft            |    |                | diagonalTopLeft       |    |                  | doubleDiamond            |    |
|         | topCenter             |   | diagonalWipe   | topLeft (default)     |    | veeWipe          | down (default)           |    |
|         | rightCenter           |  |                | leftToRight (default) |   |                  | left                     |    |
|         | bottomCenter          |  | zigZagWipe     | topToBottom           |  |                  | up                       |   |
|         | leftCenter            |  |                |                       |                                                                                       |                  | right                    |  |
|         |                       |                                                                                     | barnZigZagWipe | vertical (default)    |  |                  |                          |                                                                                       |
|         |                       |                                                                                     |                | horizontal            |  |                  |                          |                                                                                       |





# Microsoft SMIL Support

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD W3 HTML//EN">
<HTML>
  <HEAD>
    <TITLE>HTML+TIME Sample</TITLE>
    <XML:NAMESPACE PREFIX="t"/>
    <STYLE>
      /* hook up time tags */
      .time { behavior: url(#DEFAULT#TIME); }
      t\:SEQ { behavior: url(#DEFAULT#TIME); }
    </STYLE>
  </HEAD>
  <BODY>
    <t:SEQ >
      <IMG CLASS="time" t:TIMEACTION="display" SRC="p1.jpg" t:BEGIN="0"
        t:DUR="2" >
      <IMG CLASS="time" t:TIMEACTION="display" SRC="p2.jpg" t:BEGIN="0"
        t:DUR="2" >
      <IMG CLASS="time" t:TIMEACTION="display" SRC="p3.jpg" t:BEGIN="0"
        t:DUR="2" >
    </t:SEQ>
  </BODY>
</HTML>
```



# SMIL Version

---

```
<smil>
  <body>
    <seq>
      
      
      
    </seq>
  </body>
</smil>
```



# SMIL+SVG - "dynamic SVG"

---

- Based on five SMIL animation tags
  - **animate**: used to animate over time a single attribute or a single CSS property
  - **set**: provides a simple way for setting a single attribute value for a specified duration
  - **animateMotion**: causes the displacement of an element along a path of movement
  - **animateColor**: specifies a color transformation over time (deprecated – replaced by <animate> CSS color)
  - **animateTransform**: animates a transformation attribute to control translation, scaling, and rotation



# SMIL + SVG

- Used today for implementing SVG browser animation

```
<!DOCTYPE svg PUBLIC "-//W3C//DTD SVG 1.1//EN" "http://www.w3.org/Graphics/SVG/1.1/DTD/svg11.dtd">
<svg version="1.1" xmlns="http://www.w3.org/2000/svg" xmlns:xlink="http://www.w3.org/1999/xlink" width="100%"
height="100%" viewBox="0 0 1536 1024">
  <title>Demo SMIL+SVG animation</title>
  <rect x="50" y="50" height="150">
    <animate attributeName="width" fill="freeze" from="10" to="150" dur="2s" repeatCount="indefinite"/>
  </rect>
  <circle cx="50" cy="125" r="5" stroke="red" fill="#000000">
    <animateTransform
      attributeName="transform"
      begin="0s"
      dur="2s"
      type="rotate"
      from="0 125 125"
      to="360 125 125"
      repeatCount="indefinite" />
  </circle>
</svg>
```



# SMIL + SVG

- Supports interactivity through HTML events

```
<?xml version="1.0" ?>
<svg height="200" width="500" xmlns:xlink="http://www.w3.org/1999/xlink" xmlns="http://www.w3.org/2000/svg">
  <ellipse stroke-width="2" stroke="black" fill="yellow"
    cx="3cm" cy="2cm" ry="1cm" rx="2cm">
    <animate fill="freeze" dur="0.1s" to="blue" from="yellow"
      attributeName="fill" begin="mouseover"/>
    <animate fill="freeze" dur="0.1s" to="3cm"
      attributeName="rx" begin="mouseover"/>
    <animate fill="freeze" dur="0.1s" to="yellow" from="blue"
      attributeName="fill" begin="mouseout"/>
    <animate fill="freeze" dur="0.1s" to="2cm"
      attributeName="rx" begin="mouseout"/>
  </ellipse>
</svg>
```



# SMIL + SVG

- Supports animation of CSS attributes of an element

```
<?xml version="1.0" ?>
<svg height="200" width="500" xmlns:xlink="http://www.w3.org/1999/xlink" xmlns="http://www.w3.org/2000/svg">
  <ellipse stroke-width="2" stroke="black" fill="blue"
    cx="3cm" cy="2cm" ry="1cm" rx="2cm">
    <animate
      attributeType="CSS"
      attributeName="opacity"
      from="1"
      to="0"
      dur="5s"
      repeatCount="indefinite">
    </animate>
  </ellipse>
</svg>
```



# SMIL Resources

---

- W3C – The SMIL 3.0 standard

<http://www.w3.org/TR/SMIL3/>

- SMIL Tutorials

[http://edutechwiki.unige.ch/en/SVG-SMIL\\_animation\\_tutorial](http://edutechwiki.unige.ch/en/SVG-SMIL_animation_tutorial)

<https://css-tricks.com/guide-svg-animations-smil/>

- Some nice SMIL+SVG Demos

[https://upload.wikimedia.org/wikipedia/commons/9/9b/Demo\\_animated\\_feTurbulence\\_feDisplacementMap\\_SMIL.svg](https://upload.wikimedia.org/wikipedia/commons/9/9b/Demo_animated_feTurbulence_feDisplacementMap_SMIL.svg)

[https://upload.wikimedia.org/wikipedia/commons/b/ba/Toy\\_train\\_SMIL.svg](https://upload.wikimedia.org/wikipedia/commons/b/ba/Toy_train_SMIL.svg)

[https://upload.wikimedia.org/wikipedia/commons/9/9b/SMIL\\_missile\\_demo.svg](https://upload.wikimedia.org/wikipedia/commons/9/9b/SMIL_missile_demo.svg)



# SVG

---

- Scalable Vector Graphics – XML based language
- Allows defining vector images for the Web
- Zoom, transform or rotate the image without losing quality
- Image description:
  - editable with normal text editors. Smaller files
  - printed at any resolution with similar results





# SVG Features

---

- Can be generated on the fly from databases and applications
- Integrates well with server-side technology
- Objects can be grouped, styled, transformed, and reused
- Features include clipping paths, alpha masks, filter effects, template objects, and extensibility



# SVG Tags

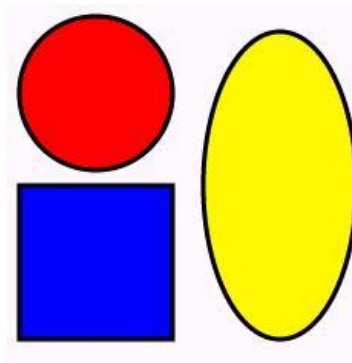
---

- **Shape:** <line>, <rect>, <circle>, <ellipse>, <polyline>, <polygon>, <path>
- **Paint:** <linearGradient>, <radialGradient>, <pattern>, <mask>
- **Text:** <text>, <tspan>, <textPath>, <font>, <font-face>, <glyph>
- **Filter:** <filter>, <feBlend>, <feFlood>, <feColorMatrix>, <feGaussianBlur>, <feMorphology>, <feSpotLight>
- **Animation:** <animate>, <animateMotion>, <animateTransform>, <animateColor>, <set>
- **Misc:** <image>, <symbol>, <g>, <defs>, <use> ...



# SVG Code

```
<svg>
<g style="stroke:black; stroke-width:3;">
  <circle cx="60" cy="60" r="50" style="fill:red" />
  <rect x="10" y="120" width="100" height="100"
    style="fill:blue" />
  <ellipse cx="180" cy="120" rx="50" ry="100"
    style="fill:yellow" />
</g>
</svg>
```





# SVG Integration

---

- **SVG file:**

```
<?xml version="1.0"?>
<svg height="200" width="500"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  xmlns="http://www.w3.org/2000/svg">
  ...
</svg>
```

- **HTML embedded:**

```
<img src = "demo.svg"/>
<svg> ... </svg>
```

- **HTML iframe:**

```
<iframe src="demo.svg" width="300" height="100">
</iframe>
```



# SVG Resources

---

- W3C SVG 1.1

<http://www.w3.org/TR/SVG11/>

- SVG tutorial (w3schools)

[https://www.w3schools.com/graphics/svg\\_intro.asp](https://www.w3schools.com/graphics/svg_intro.asp)

- Pike's SVG Tutorial

[http://apike.ca/prog\\_svg.html](http://apike.ca/prog_svg.html)

- Examples

<https://www.paulirish.com/2010/svg-filters-on-html5-video/>